OpenAI HR Attrition Insights Generator

Capstone Documentation and Technical Toolkit

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Project focus	AI-assisted HR attrition analysis
Primary technology	OpenAI API with Python
Document purpose	Guide, technical record, and portfolio documentation

Executive Summary

This project documents the design and implementation of an AI-powered HR attrition insights generator built with Python, pandas, and the OpenAI API. The tool reads a structured HR attrition dataset, performs data cleaning and summary analysis, and then uses a large language model to transform numerical outputs into clear, business-ready insights.

The work was intentionally designed as a beginner-friendly capstone: the workflow is simple enough to replicate in Google Colab, yet strong enough to demonstrate practical data skills such as feature engineering, prompt design, troubleshooting, documentation, and output interpretation.

The final system answers a real business need. Instead of handing decision-makers a spreadsheet or a block of statistics, it generates concise insight statements, likely causes of attrition, recommended actions, and a summary of business risk. This demonstrates how generative AI can extend the value of traditional analytics rather than replace it.

Document Map

- Project purpose and chosen technology
- System requirements and setup instructions
- Implementation workflow and minimal working example
- AI prompt journal and iteration record
- Common issues, fixes, and lessons learned
- Testing, business value, future improvements, and conclusion

1. Project Title and Objective

Project title	OpenAI HR Attrition Insights Generator
Objective	To build a beginner-friendly tool that analyzes an HR attrition dataset and uses the OpenAI API to generate human-readable, decision-oriented business insights.
End goal	Produce a repeatable workflow in which structured HR metrics are converted into concise findings, likely root causes, recommended interventions, and an explicit business risk statement.

2. Overview of the Chosen Technology

The primary technology used in this project is the OpenAI API, supported by Python and the pandas library. The OpenAI API is an application programming interface that allows developers to send prompts to

advanced language models and receive generated outputs such as summaries, explanations, classifications, and recommendations.

Python was selected because it is one of the most widely used languages in analytics and data science. It is well suited for reading files, cleaning data, creating derived variables, calculating summary statistics, and integrating with web-based services such as APIs. pandas was chosen because it provides a fast and readable way to manipulate tabular data.

This technology combination was chosen for two reasons. First, it aligns directly with the capstone requirement to learn a new technology using generative AI. Second, it solves a practical analytics problem: many stakeholders struggle to interpret raw statistics, so the project uses AI to turn numerical summaries into plain-language business insights.

Why this stack is relevant

Component	Role in project	Why it matters
Python	Loads files, cleans text fields, creates derived features, and computes summary metrics	Provides the analytical backbone of the workflow
pandas	Groups, transforms, and summarizes HR data	Makes the workflow concise, reproducible, and easy to explain
OpenAI API	Converts summary statistics into insight statements and recommendations	Shows how generative AI can augment traditional analytics

3. System Requirements

Requirement	Details
Operating environment	Google Colab or Jupyter Notebook
Python version	Python 3.10 or newer
Internet access	Required for API requests
OpenAI account	Required to create an API key and use model access
Core libraries	openai, pandas, openpyxl

Installation command

```
pip install openai pandas openpyxl
```

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```

4. Installation and Setup Workflow

Step 1 - Prepare the notebook environment

Install the required libraries in Google Colab and confirm that pandas and the OpenAI library are available.

Step 2 - Upload the HR dataset

Load the Excel file into Colab and inspect the file name, shape, and columns to confirm that the dataset has been read correctly.

Step 3 - Enter the API key securely

Use a hidden input method rather than hardcoding the key in a notebook cell. This protects sensitive credentials.

Step 4 - Run the data preparation pipeline

Clean column names, standardize text categories, create binary attrition labels, and define age bands.

Step 5 - Generate the analytical summary

Compute overall attrition, department attrition, job-role attrition, age-group attrition, and income differences between leavers and stayers.

Step 6 - Send the summary to OpenAI

Use a structured prompt to request key insights, likely causes, recommendations, and risk interpretation.

Step 7 - Save and document the output

Store the generated insights in a text file and capture screenshots for GitHub and the final report.

5. Data Processing and Feature Engineering

A large portion of the work involved preparing the dataset before sending anything to the language model. This was essential because the quality of AI-generated insight depends heavily on the quality of the analytical summary used as input.

The dataset initially contained duplicate-looking department categories caused by formatting differences, such as changes in capitalization. If these were not corrected, the grouped statistics would produce false extra categories and distort the attrition results.

Cleaning logic used

```
df["Department"] = df["Department"].astype(str).str.strip().str.lower()
df["Department"] = df["Department"].replace({
    "human resources": "Human Resources",
    "sales": "Sales",
    "research & development": "Research & Development"
})
df["Attrition_Binary"] = df["Attrition"].map({"Yes": 1, "No": 0})
```

```
df["AgeGroup"] = pd.cut(df["Age"], bins=[18, 30, 40, 50, 60],
                        labels=["18-30", "31-40", "41-50", "51-60"])
```

Derived variables

- Attrition_Binary: converts the Attrition field from Yes/No into 1/0 so that means and rates can be calculated easily.
- AgeGroup: segments employees into interpretable age bands to support pattern analysis rather than relying only on raw age values.

6. Analytical Summary Produced Before the API Call

Before sending any request to the API, the notebook generated a structured summary containing the key facts the model needed to interpret. This kept the prompt grounded in real calculations rather than asking the model to infer directly from a raw file.

The summary included: total employees, overall attrition rate, department attrition rates, top job roles with attrition, age-group attrition, and average monthly income for leavers versus employees who stayed.

Metrics included in the summary

Metric	Why it was included
Overall attrition rate	A baseline measure of the size of the retention problem.
Department attrition	Helps identify which business areas are experiencing the highest losses.
Job-role attrition	Surfaces concentrated turnover risk in specific roles.
Age-group attrition	Supports segmentation and may reveal early-career or mid-career risk patterns.
Income comparison	Highlights whether compensation differences may be associated with attrition.

7. Minimal Working Example

The minimal working example for this capstone is a notebook-based pipeline that:

1. Loads an Excel HR attrition dataset.
2. Cleans the data and creates a few derived features.
3. Calculates grouped attrition statistics.
4. Builds a summary string from the computed results.

5. Sends that summary to the OpenAI API.
6. Receives a formatted response containing insights, root causes, recommendations, and a business risk statement.

Sample API prompt structure

```
You are a senior HR analytics consultant.

Based on the HR attrition summary below:
1. Provide 10 key insights that are specific and data-driven
2. Identify 5 likely root causes of attrition
3. Provide 3 actionable recommendations
4. Highlight 1 key business risk if attrition is not addressed

Keep the tone professional, concise, and suitable for a management report.

{summary}
```

8. AI Prompt Journal and Iteration Record

One of the strongest lessons in this project was that prompt quality directly influenced output quality. Early prompts produced generic observations, while more specific prompts generated structured and useful business commentary.

Version	Prompt approach	Observed issue / improvement	Lesson learned
Prompt 1	Generate insights from HR attrition summary	Output was too general and lacked business structure	Broad prompts can work, but they often miss depth and specificity
Prompt 2	Assigned the model a role: senior HR analytics consultant	Tone became more professional and focused	Role prompting helps shape voice and domain focus
Prompt 3	Added numbered requirements: insights, root causes, recommendations, risk	Response became organized and easier to use in reporting	Explicit output structure improves usefulness
Prompt 4	Improved the analytical summary by adding age-group attrition	The model had more context for segmentation-based commentary	Better inputs lead to better AI outputs

9. Common Issues Encountered and How They Were Solved

Duplicate department categories

Cause: Different capitalizations created false extra departments after grouping.

Fix: Standardized department text using strip, lower, and explicit replacement mapping.

Syntax error while printing age-group output

Cause: Text and curly braces were typed outside an f-string or print function.

Fix: Used print statements or inserted the text inside the summary f-string.

API quota error

Cause: The API call initially returned a 429 insufficient_quota error.

Fix: Reviewed billing/quota setup, regenerated the API key, and retried with a valid project configuration.

GitHub folder confusion

Cause: A file named Screenshots was created instead of a folder, which blocked file moves.

Fix: Deleted the file and recreated the folder properly using the screenshots/.gitkeep pattern.

10. Testing and Validation

The project was tested in stages rather than only at the end. This made troubleshooting easier and helped confirm that each part of the workflow behaved as expected before moving on to the next one.

- Dataset load test: verified that the Excel file opened correctly and displayed the expected columns.
- Cleaning validation: checked unique department values to confirm that duplicates were removed.
- Feature validation: confirmed that Attrition_Binary and AgeGroup were created correctly.
- Summary test: printed the summary string and reviewed whether all required metrics appeared correctly.
- API response test: verified that the OpenAI call returned a structured response with all requested sections.
- Output persistence test: saved generated insights to a text file so results could be reused and uploaded to GitHub.

11. Business Value and Practical Relevance

The project has clear business value because employee attrition affects continuity, recruitment cost, training cost, workload balance, and institutional knowledge. Traditional dashboards can show where attrition is happening, but they do not always communicate the meaning of the numbers effectively to non-technical users.

By turning structured metrics into narrative insight, the tool helps managers and analysts move more quickly from observation to action. It can support HR reporting, department-level reviews, executive summaries, and early discussion of retention strategies.

Potential organizational benefits

- Identify high-risk departments and job roles.
- Surface compensation-related attrition patterns.
- Improve the communication of analytics to managers and executives.
- Reduce the time needed to convert raw analysis into written commentary.
- Create a repeatable starting point for more advanced HR decision-support systems.

12. Documentation, GitHub Packaging, and Portfolio Readiness

The project was also documented for GitHub so that it could function both as a capstone submission and as a portfolio asset. The repository contains a README file, a toolkit document, requirements, screenshots, and saved sample outputs.

This packaging step matters because good technical work is more useful when it is reproducible, readable, and easy for another person to evaluate. A strong repository shows not just coding ability, but also professionalism and communication skill.

Repository structure

```
openai-hr-attrition-insights-generator/
|- README.md
|- toolkit.md
|- requirements.txt
|- hr_attrition_insights_generator.ipynb
|- screenshots/
|  |- hr_data.png
|  |- ai_insights_output.png
|- sample_output/
|  |- attrition_insights.txt
```

13. Future Improvements

- Streamlit application: convert the notebook workflow into an interactive web app for portfolio presentation.
- Predictive modeling: add a statistical or machine learning component for attrition risk scoring.
- Dashboard integration: pair the written insights with visual summaries in Power BI or a web dashboard.
- Richer HR features: incorporate overtime, years at company, satisfaction metrics, and other potential drivers.
- Business adaptation: extend the workflow to other operations such as hospitality, finance, or NGO monitoring and evaluation.

14. Reflection and Key Learning

This capstone demonstrated that working with generative AI is not just about sending a prompt to a model. Strong results depend on the quality of the full workflow: data preparation, metric selection, prompt design, validation, and documentation.

It also reinforced an important analytics principle: numbers become more valuable when they are translated into meaning. The OpenAI API did not replace the analysis; instead, it extended the analysis by helping present it in a form that is easier for stakeholders to understand and act on.

A final lesson was that iteration matters. Several project improvements came from debugging, testing different prompt structures, correcting category duplication, and packaging the work carefully on GitHub. Each of those steps strengthened both the project itself and the quality of the final submission.

15. References

- OpenAI API documentation
- pandas documentation
- Google Colab documentation
- Python documentation

Conclusion

The OpenAI HR Attrition Insights Generator successfully shows how generative AI can be integrated into a practical analytics workflow. By combining structured data analysis with language-model-based interpretation, the project produces outputs that are more accessible, decision-ready, and reusable than raw statistics alone.

As both a capstone submission and a portfolio artifact, the project demonstrates technical curiosity, analytical thinking, prompt engineering, debugging, and clear communication. It also lays the groundwork for future AI-assisted systems, including more advanced decision-support tools and domain-specific agents.